Muhammet Celal Gokalp

Solutions Engineer | AI-Enabled Engineering Workflows | Autonomous Systems

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PROFESSIONAL SUMMARY

Customer-facing engineer with nearly 8 years in automotive and autonomous driving, embedded with OEM and internal customer teams to turn messy requirements into working, verified solutions. At FiveAI / Bosch I built LLM-based agent skills with GitHub Copilot and Copilot CLI that turned regulation and Euro NCAP rules into OpenSCENARIO 2 test scenarios. The approach was adopted across the project and multiplied scenario output from a small set of hand-crafted modular building blocks. I am not a classically trained software engineer. I learned to build and ship tools by working closely with AI, which is why I am good at showing non-specialists how to use it well, with healthy scepticism and careful checking of results.

WORK EXPERIENCE

Lead Application Engineer (Solutions & Integration)

10/2021 – Present

FiveAI / Bosch — Cambridge, United Kingdom

- **LLM agent skills for scenario generation:** Built AI skills with GitHub Copilot and Copilot CLI that act as an OpenSCENARIO 2 (DSL) helper. Engineers give the agent context such as UN regulation text and Euro NCAP rules, and it composes our hand-crafted modular scenarios into new ones with exact, checkable metrics. Used project-wide, it increased scenario output by an order of magnitude.
- **Driving AI adoption:** Showed engineers and non-software colleagues how to use AI tools in their daily work, including how to give good context, break problems down, and verify outputs rather than trust them.
- **Domain-specific evaluation:** Every generated scenario was checked against its source rules and run in simulation. Defined pass/fail and KPI criteria, such as comfort limits for acceleration and deceleration across 1,000+ hours of test data.
- **Scenario-based V&V:** Built 120+ parameterised OpenSCENARIO 2 scenarios generating thousands of test cases for an L4 automated driving stack, covering Euro NCAP junction tests and UN R79 automatic lane change (ADA project).
- **Tools & data pipelines:** With AI-assisted coding in Python, built internal tools (a GUI for JSON test parameters, a local uploader) and pipelines that send simulation results to front-end dashboards, raising pull requests to the main monorepo and working in its dev container.
- **Internal customers & integration:** On the ADA project, a partnership with Volkswagen Group where FiveAI supplied the simulation tooling, trained partner engineers to use our toolchain. This grew into the Simulation EcoSystem solutions team, where I was a founding member. Ran discovery with users, triaged issues across tools, maps and simulation, and worked with several engineering teams to keep internal and enterprise users running.

ADAS Systems Engineer (Feature Owner, client-embedded)

11/2018 – 10/2021

AVL Powertrain UK — Coventry, United Kingdom

- **Embedded at the customer:** Worked on-site at Jaguar Land Rover for two years as AVL's feature owner for Rear Cross Traffic Alert, Blind Spot Monitor and Rear Pre-Crash. I was the technical link between the OEM, AVL and suppliers.
- **Discovery to delivery:** Turned stakeholder needs into use cases, system state diagrams and requirements in DOORS and MKS Integrity (ASPICE), then verified them on HiL and ViL rigs and on the proving ground.
- **Problem solving under ambiguity:** Found root causes in vehicle network data (Vector CANoe / CANape). For example, an alert triggered late at boundary conditions, and I agreed a sensor field-of-view change with the supplier while keeping false alerts low.
- **Safety mindset:** Wrote ISO 26262 functional safety requirements for a lane change feature. Trained in ISO 26262 and SOTIF (ISO 21448).

Part-Time Teacher and Program Planner

12/2013 – 01/2017

Space Camp Turkey — Izmir, Türkiye

- Taught space technology and science to students aged 9 to 16. This is where I learned to explain complex ideas simply.

EDUCATION

MSc in Sustainable Automotive Engineering

10/2017 – 09/2018

Warwick Manufacturing Group, Coventry, UK

- **Grade:** 2:1 (Merit). Dissertation: experimental testing to find the variables of a thermal model of an electric motor stator (MATLAB, Simulink, MotorCAD).

BSc in Mechanical Engineering

08/2011 – 02/2017

Ege University, Izmir, Türkiye

- Dissertation: designed, built and controlled a 6-DOF Stewart motion platform (SolidWorks, MATLAB, Arduino).

SKILLS

AI & LLMs	Agent skills and prompting (GitHub Copilot, Copilot CLI), giving domain context to LLMs, AI-assisted coding, checking LLM output against rules and simulation
Software	Python (scripting, tools, data pipelines), C++ (reading), OpenSCENARIO 2 DSL, Git / GitHub, monorepo and dev containers, JSON, MATLAB / Simulink
Customer & Delivery	On-site client work, requirements discovery, stakeholder management, root cause analysis, training colleagues, Jira / Confluence
Domain	L4 autonomous driving, ADAS, scenario-based testing, SIL / HiL / ViL, ISO 26262, SOTIF, UN R79, Euro NCAP
Languages	English (C2), Turkish (Native)